

TOPIC :

Conditional Probability.

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Conditional Probability Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B
Probability of A and B
Probability of B

Similar

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

means - B ho chukaye
A → hone ka probability.

if A & B are independent event

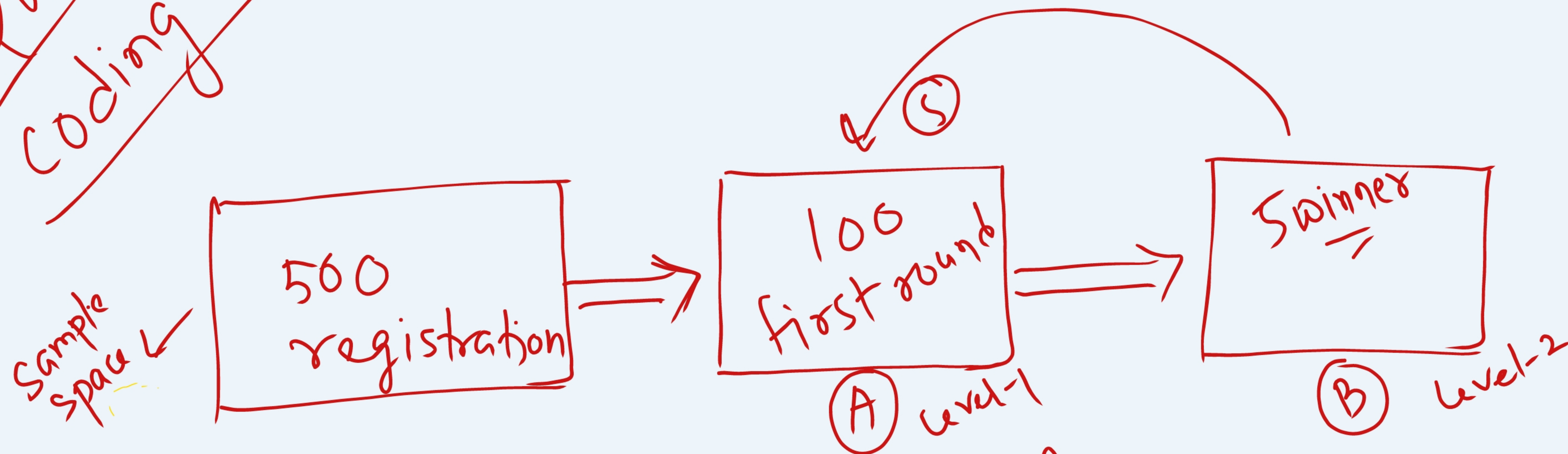
$$P(A \cap B) = P(A) \cdot P(B) \checkmark$$

$$\therefore P(A|B) = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

$$\therefore P(B|A) = P(B) \checkmark$$



QuadAI Coding Competition



what is the Conditional Probability of event B when A has Already happened?

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{5}{100} = \frac{1}{20}$$

$$P(A) = \frac{100}{500} = \frac{1}{5}$$

$$P(B) = \frac{5}{500} = \frac{1}{100}$$

$$P(B|A) = \frac{5}{100} = \frac{1}{20}$$

conditional probability of

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{5}{100} = \frac{1}{20}$$

Prob-1

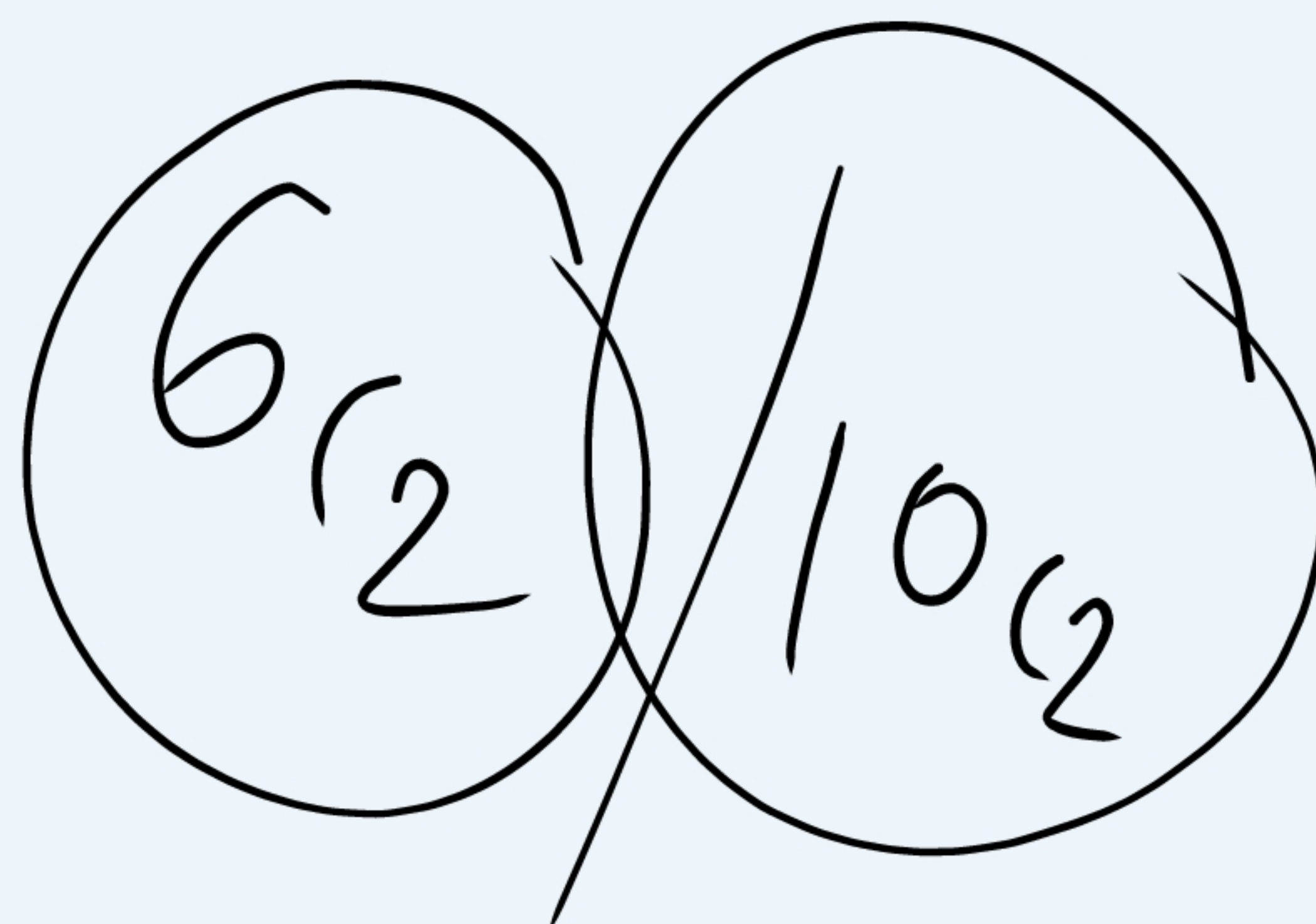
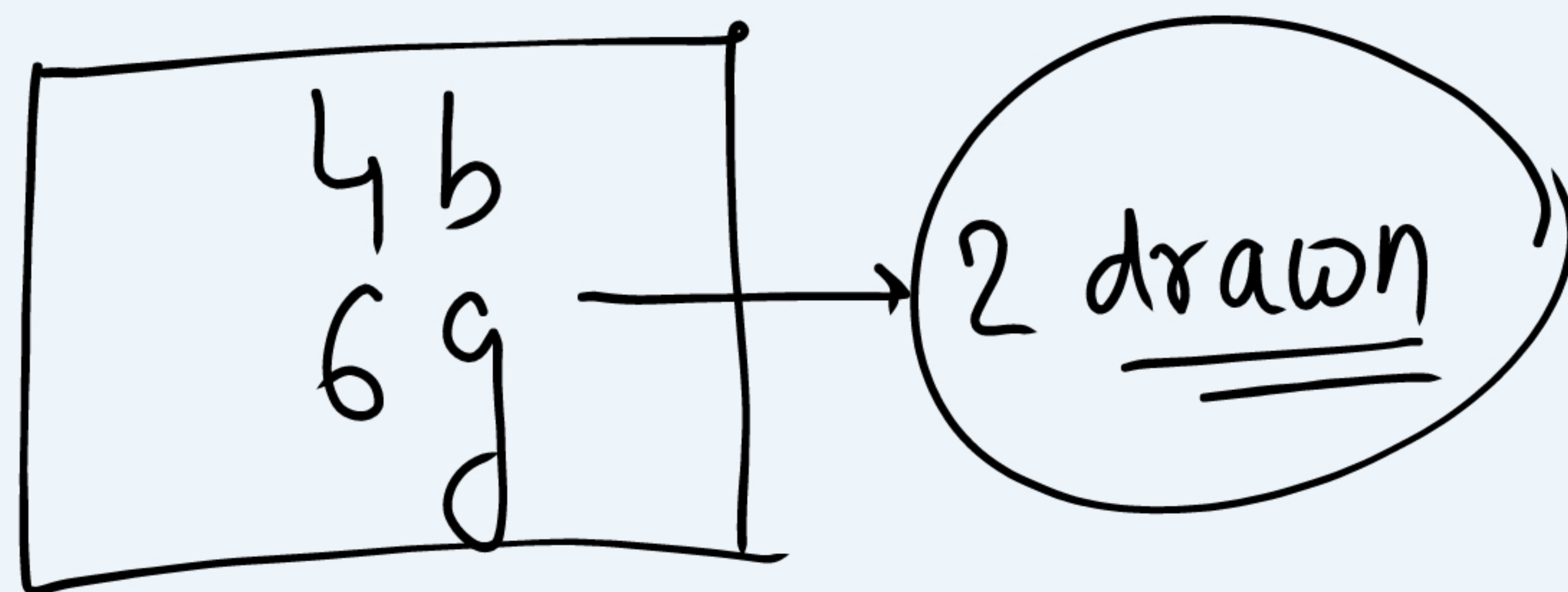
if an odd number comes up on tossing a die. Find the Probability of its being Prime number?

$S = \{1, 2, 3, 4, 5, 6\}$
 $A = \{1, 3, 5\} \rightarrow \text{odd}$
 $B = \{2, 3, 5\} \rightarrow \text{prime}$
 $A \cap B = \{3, 5\}$

$$\begin{aligned} \therefore P(B/A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{2/6}{3/6} = \left(\frac{2}{3}\right) \text{ Ans} \checkmark \end{aligned}$$

Prob 2

A box contains 4 bad and 6 good tubes. Two are drawn out from the box at a time. One of them is tested and found to be good. What is the prob. that the other one is also good?



$$\begin{aligned}
 P(G/G) &= \frac{P(G \cap G)}{P(G)} \\
 &= \frac{6C2}{6C1 \cdot 10C1} \\
 &= \frac{6 \times 5}{6 \times 10} \\
 &= \frac{10 \times 5}{10 \times 6} \\
 &= \frac{5}{6} \checkmark
 \end{aligned}$$

Prob-3

Let, $P(E)$ denotes the probability of the independent event E .

Given, $P(A) = 1$
 $P(B) = \frac{1}{2}$ ✓
the value of $P(A/B)$ and $P(B/A)$ respectively are -

- Ⓐ $\frac{1}{4}, \frac{1}{2}$
- Ⓑ $\frac{1}{2}, 1$
- Ⓒ $\frac{1}{2}, \frac{1}{4}$
- Ⓓ $1, \frac{1}{2}$ ✓

$$P(A \cap B) = P(A) \cdot P(B) \\ = 1 \cdot \frac{1}{2}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$= 1 \\ P(B/A) = \frac{P(B \cap A)}{P(A)} = \left(\frac{1}{2} \right)$$

Prob-4

A hydraulic structure has four gates which operate independently. The probability of failure of each gate is 0.2 and Given that 1 has failed, the probability that both gates 2 and 3 will fail is -

- (a) 0.240
- (b) 0.200
- (c) 0.040
- (d) 0.008 ✓

$$\begin{aligned} & P(\text{gate 2 and 3 failed} \mid \text{1 has failed}) \\ &= \frac{P(\text{gate 2 and gate 3 fail and gate 1 fail})}{P(\text{gate 1 failed})} \\ &= \frac{0.2 \times 0.2 \times 0.2}{0.2} \\ &= 0.04 \checkmark \end{aligned}$$

Problem 1.

The chance of a student passing an exam is 20%. The chance of a student passing the exam and getting above 90% marks is 5%.

Given that a student passes the examination, the probability that the student gets above 90% marks is -

- (a) $\frac{1}{18}$
(b) $\frac{1}{4}$
(c) $\frac{2}{9}$
(d) $\frac{5}{18}$

90% marks is -

$$P(90\% \text{ marks}) \Rightarrow \frac{P(90\% \text{ marks})}{P(\text{Passing an exam})} \Rightarrow \frac{5}{20}$$
$$\Rightarrow \frac{1}{4}$$

$$n(A) = 70$$
$$n(B) = 60$$
$$n(A \cap B) = 40$$

$$\frac{40}{60} = \frac{2}{3}$$

Prob.

A family has two children. What is the probability that both children are boys given that at least one of them is boy?

④ $\frac{1}{2}$
⑤ $\frac{1}{3}$
⑥ $\frac{1}{5}$
⑦ $\frac{1}{4}$

(g, b)
 (b, g)

let, b represents boy
and g represents girl.

$S = \{ \textcircled{bb}, \cancel{bg}, \textcircled{gg}, gb \} = 4$

$E =$ both the children are boys
 $F =$ at least one of them is a boy

$$\therefore P(E/F) = \frac{P(E \cap F)}{P(F)}$$

$$= \frac{\frac{1}{4}}{\frac{3}{4}} = \textcircled{\frac{1}{3}}$$

$$E = \{ \textcircled{bb} \}$$

$$F = \{ bb, bg, gb \}$$

$$E \cap F = \{ \textcircled{bb} \} = \frac{1}{4}$$